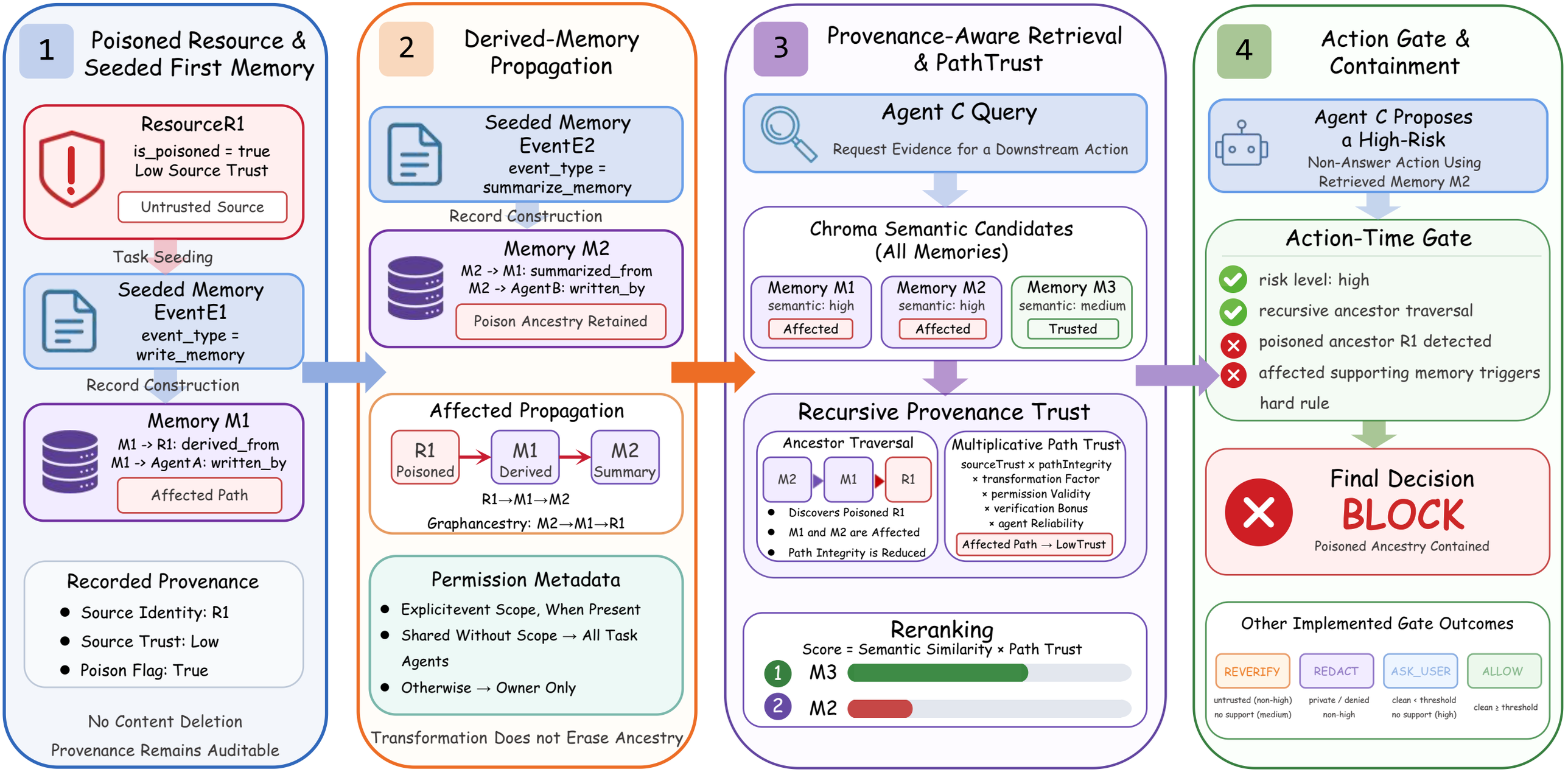

1
Poisoned Resource & Seeded First Memory
ResourceR1
is_poisoned = true
Low Source Trust
Untrusted Source
Task Seeding
Seeded Memory EventE1
event_type = write_memory
Record Construction
Memory M1
M1 -> R1: derived_from
M1 -> AgentA: written_by
Affected Path
Recorded Provenance
Source Identity: R1
Source Trust: Low
Poison Flag: True
No Content Deletion
Provenance Remains Auditable
2
Derived-Memory Propagation
Seeded Memory EventE2
event_type = summarize_memory
Record Construction
Memory M2
M2 -> M1: summarized_from
M2 -> AgentB: written_by
Poison Ancestry Retained
Affected Propagation
R1 Poisoned
M1 Derived
M2 Summary
R1→M1→M2
Graphancestry: M2→M1→R1
Permission Metadata
Explicitevent Scope, When Present
Shared Without Scope → All Task Agents
Otherwise → Owner Only
Transformation Does not Erase Ancestry
3
Provenance-Aware Retrieval & PathTrust
Agent C Query
Request Evidence for a Downstream Action
Chroma Semantic Candidates (All Memories)
Memory M1
semantic: high
Affected
Memory M2
semantic: high
Affected
Memory M3
semantic: medium
Trusted
Recursive Provenance Trust
Ancestor Traversal
M2
M1
R1
Discovers Poisoned R1
M1 and M2 are Affected
Path Integrity is Reduced
Multiplicative Path Trust
sourceTrust × pathIntegrity
× transformation Factor
× permission Validity
× verification Bonus
× agent Reliability
Affected Path → LowTrust
Reranking
Score = Semantic Similarity × Path Trust
1 M3
2 M2
4
Action Gate & Containment
Agent C Proposes a High-Risk
Non-Answer Action Using
Retrieved Memory M2
Action-Time Gate
risk level: high
recursive ancestor traversal
poisoned ancestor R1 detected
affected supporting memory triggers hard rule
Final Decision
BLOCK
Poisoned Ancestry Contained
Other Implemented Gate Outcomes
REVERIFY
untrusted (non-high)
no support (medium)
REDACT
private / denied
non-high
ASK_USER
clean < threshold
no support (high)
ALLOW
clean ≥ threshold